Scientific Calculator Project

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Description:

This document is prepared for the Scientific Calculator project in Python.

It includes GUI implementation using Tkinter and modern styling options.

Details:

- Language: Python

- GUI Framework: Tkinter

- Features: Arithmetic operations, scientific functions (sqrt, sin, cos, tan, log)

- Modern Styling: Dark theme, colored buttons

CODE:

```
import tkinter as tk
import math

# Create main window
root = tk.Tk()
root.title("Scientific Calculator")
root.configure(bg="#1e1e1e")  # Dark background

# Entry widget for input/output
entry = tk.Entry(root, width=30, borderwidth=5,
                 font=("Arial", 16, "bold"),
                 bg="#2d2d2d", fg="white", insertbackground="white")
entry.grid(row=0, column=0, columnspan=5, padx=10, pady=10)

# Function to insert values
def click_button(value):
    entry.insert(tk.END, value)
```

```
# Function to clear entry
def clear():
    entry.delete(0, tk.END)

# Function to evaluate expression
def calculate():
    try:
        result = eval(entry.get())
        entry.delete(0, tk.END)
        entry.insert(0, result)
    except Exception:
        entry.delete(0, tk.END)
        entry.insert(0, "Error")

# Function for scientific operations
def scientific(func):
    try:
        value = float(entry.get())
        if func == "sqrt":
            result = math.sqrt(value)
        elif func == "sin":
            result = math.sin(math.radians(value))
        elif func == "cos":
            result = math.cos(math.radians(value))
        elif func == "tan":
            result = math.tan(math.radians(value))
```

```
        elif func == "log":
            result = math.log10(value)
        entry.delete(0, tk.END)
        entry.insert(0, result)
    except Exception:
        entry.delete(0, tk.END)
        entry.insert(0, "Error")

# Buttons layout
buttons = [
    ('7',1,0), ('8',1,1), ('9',1,2), ('/',1,3), ('sqrt',1,4),
    ('4',2,0), ('5',2,1), ('6',2,2), ('*',2,3), ('sin',2,4),
    ('1',3,0), ('2',3,1), ('3',3,2), ('-',3,3), ('cos',3,4),
    ('0',4,0), ('.',4,1), ('+',4,2), ('=',4,3), ('tan',4,4),
    ('C',5,0), ('log',5,1)
]

for (text, row, col) in buttons:
    if text == "=":
        tk.Button(root, text=text, width=6, height=2,
                  bg="#4CAF50", fg="white", font=("Arial", 12, "bold"),
                  command=calculate).grid(row=row, column=col, padx=5, pady=5)
    elif text == "C":
        tk.Button(root, text=text, width=6, height=2,
                  bg="#f44336", fg="white", font=("Arial", 12, "bold"),
                  command=clear).grid(row=row, column=col, padx=5, pady=5)
    elif text in ["sqrt","sin","cos","tan","log"]:
```

```
        tk.Button(root, text=text, width=6, height=2,
                  bg="#2196F3", fg="white", font=("Arial", 12),
                  command=lambda t=text: scientific(t)).grid(row=row, column=col, padx=5, pady=5)
    else:
        tk.Button(root, text=text, width=6, height=2,
                  bg="#333333", fg="white", font=("Arial", 12),
                  command=lambda t=text: click_button(t)).grid(row=row, column=col, padx=5,
pady=5)


root.mainloop()
```